

Recod.ai/LUC - Scientific Image Forgery Detection

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Abstract

This paper presents and compares a range of DINOv2-based model architectures for the Recod.AI Scientific Image Forgery Detection Kaggle competition. The comparison highlights different useful techniques for Copy-Move Forgery Detection (CMFD) as well as dataset/task-specific considerations when performing CMFD. Although we were unable to test our method on the Kaggle test set due to late entry, we achieve a holdout oF1 of 0.5138 on an internal stratified holdout test set derived from the Kaggle training data using a model combining DINOv2 features with BusterNet (BusterDino).

Additionally, we examine the role of strong pretrained feature extraction for this task by implementing a SegNeXt-inspired segmentation baseline and comparing it with the project’s DINO-based baseline. The results indicate that strong pretrained features are important for CMFD in this setting.

1 Introduction

1.1 Copy-Move Forgery Detection

Copy-Move Forgery is a class of image forgery where regions of an image are duplicated and moved to other areas of the image. Commonly, techniques such as rotation, flipping and scaling of the copied region are applied, as well as post-processing such as edge smoothing, brightness adjustment and adding extra noise. These techniques can lead to forgeries which are hard to detect both through manual review and automated detectors. This method of forgery can be used within scientific images in order to fabricate results, where the complexity of the figures creates additional challenges. [1]

1.2 Related works

Copy-move image forgery detection has been studied using a range of techniques. Early approaches relied on hand-crafted descriptors with nearest-neighbour matching and geometric consistency checks [2, 3], but are sensitive to strong post-processing, geometric transformations, and repeated structures [3].

More recent work focuses on deep learning. Some methods retain the matching paradigm with learned features [4], while others formulate the task as end-to-end forgery localization using CNN-based architectures [5]. Attention-based and transformer-based models further improve performance by modelling patch relationships and global context [6, 7]. Source-target disambiguation has also emerged as an important subproblem [8].

Some methods seek to combine the strengths of both approaches, such as one of the architectures examined in this paper which combines learned CNN features with Zernike-based descriptors [9].

1.3 Objectives

We aim to create a full prediction pipeline for CMFD which could perform well on the Recod.AI Scientific Image Forgery Detection dataset. Ideally, our model should be robust to post-processing techniques. The main goal is to perform well on the competition metric, which is a modified version of F1 which computes pixelwise F1 on pairs of (*predicted mask*, *ground truth mask*) using the following formula [10]:

$$F1 = \frac{2TP}{2TP + FP + FN}$$

This calculation is applied to every connected component of forged pixels in the image (belonging to either the source or target). Predictions given by the model are matched up with the best corresponding ground-truth areas using the Hungarian algorithm. When the number of predicted components exceeds the number of ground-truth components, a penalty is computed:

$$P = \frac{n_{gt}}{\max(n_{pred}, n_{gt})}$$

Finally, the per-image score is given as:

$$oF1 = \text{mean}(\text{best matched pairwise F1 scores}) \times P$$

The competition metric is challenging for a few reasons. Firstly, it requires precise pixel-level localization for the output masks. In addition, a score of 0 for a given image is given if any pixels are predicted as forged on a ground-truth authentic image. Finally, it is highly punishing if any additional components are predicted, even if they should be small,

as this affects the penalty P . These factors mean that the metric particularly encourages low false positive rate for authentic samples.

1.4 Contributions

This paper presents several complete pipelines for CMFD on the Recod.AI Scientific Image Forgery Detection dataset. The main CMFD pipelines are built around DINOv2 features [11], where the baseline model passes the features directly through a decoder. To build on this, we also present more complex architectures based on BusterNet [5] and Cross-Scale PatchMatch [9]. The strengths and weaknesses of these approaches are discussed, providing an empirical basis for comparing different design choices. Through this analysis, the paper identifies directions that appear promising for future work and further nuances the task of CMFD.

2 Methods

We run two experiments. Firstly, our main experiment assesses the performance of the different methods based on DINOv2 features. All experiments were conducted with a fixed random seed of 42. We also seeded Python, NumPy, PyTorch, CUDA, and data-loader workers where applicable, and disabled cuDNN benchmarking to make the runs highly reproducible. However, we do not claim strict bit-wise determinism, as some PyTorch/CUDA operations and architecture-dependent kernels can remain nondeterministic in practice, and enforcing fully deterministic algorithms would have made parts of the training pipeline impractical.

Our secondary experiment evaluates feature extraction strength of DINOv2, by comparing it against a pretrained SegNeXt-inspired model for different dataset sizes, measuring validation F1, without applying any post-processing in order to keep this experiment clean.

2.1 Dataset

We utilize the dataset supplied in the Kaggle competition [1]. Unfortunately, we were unable to test our implementation against the Kaggle test-set, as we were too late to enter the competition. In our main experiment, we chose to use the available training data in a train (80%), validation (10%), test (10%) split. In our secondary experiment, we use different-sized subsets of the same dataset in order to assess the impact of dataset size against feature extraction. The training data consists of a base 2751 forged and 2377 authentic images as well as an additional 48 forged images. As the images are too large to process directly with the model, they are each resized

to size $3 \times 448 \times 448$. The corresponding masks are resized using nearest-neighbour interpolation to preserve binary labels.

2.2 Training Objective and Optimization

All trainable prediction heads are optimized as pixel-wise binary classification problems, where each pixel is classified as either forged or authentic. We use the base binary cross-entropy loss for pixel-wise supervision, and utilize the Adam optimizer.

$$\mathcal{L}_{BCE} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)]$$

Some architectures use additional loss terms, described in their respective sections.

2.3 Baseline Model

As the Kaggle competition had already been running for 3 months when we began our experiments, our initial implementation was inspired by the techniques used by other users [12, 13]. We noted that many users employed the DINOv2 model [11], as well as techniques for prediction mask post-processing. Our baseline model is inspired by the use of these techniques.

2.3.1 Feature Extraction

This model first extracts general-purpose features using the frozen DINOv2 ViT backbone. For a 448×448 input, the ViT-B/14 backbone produces a 32×32 grid of patch embeddings with dimension 768, which is then passed to the decoder for mask prediction.

2.3.2 Decoder

To generate the predicted mask, we use a lightweight convolutional decoder on top of the DINOv2 feature map. The decoder consists of three convolutional blocks followed by an output layer, and produces a mask upsampled to the input resolution using bilinear interpolation.

The feature dimensionality is reduced progressively from $768 \rightarrow 384 \rightarrow 192 \rightarrow 96$, with ReLU activations and Dropout ($p = 0.1$) applied in each block, before a final $96 \rightarrow 1$ projection.

2.4 BusterDino

Our modern implementation of BusterNet [5] uses the concepts from the paper of a similarity layer, a two-branch approach with a fusion model, and a 3 stage training setup. Wu et al. emphasize that

the core concepts of BusterNet are most central and that inner implementations could be changed by the reader. This inspired our choices to adapt the inner architecture to work with the DINOv2 model as a feature extractor, and to utilize modern deep learning principles for efficiency and robustness. For more implementation details, see GitHub [14].

2.5 Deep Dino Cross-Scale Patch-Match

The Deep Cross-Scale PatchMatch architecture [9] combines two branches: a PatchMatch-based module [15] that estimates pixel correspondences across scales, and a direct feature extraction $\rightarrow$ prediction branch. Their outputs are combined via an argmax operation.

We adapt the architecture to single-channel CMFD and replace the original feature extractors with frozen pretrained CNNs, avoiding back-propagation through the PatchMatch module and reducing computational cost. In the first branch, we use the less computationally intensive ResNet18 backbone [16] single-scale in tandem with multi-scale Zernike feature extraction. In the second branch, we utilize DINOv2 for feature extraction followed by a similar decoder to our baseline method.

In addition to the original dice losses for the final mask M and SEUnet output M_t , we include the binary cross-entropy used in the baseline loss as well as two auxiliary terms: (i) a dice loss on the PatchMatch output M' to prevent collapse, and (ii) a penalty on predictions for authentic images to reduce false positives. These are weighted by 0.8 and 0.25, respectively. For more implementation details, see GitHub [17].

2.6 SegNeXt-inspired implementation

SegNeXt is motivated by the idea that semantic segmentation benefits from strong encoders, multi-scale feature interaction, and efficient spatial attention implemented with convolutional operations rather than transformer self-attention [18]. It employs a convolutional encoder-decoder architecture with multi-scale convolutional attention to capture contextual information efficiently, achieving strong performance-computation trade-offs and outperforming transformer-based alternatives on several benchmarks [18]. We use SegNeXt to evaluate whether a lightweight CNN-based approach can match the stronger DINO-based baseline.

Our goal was not to reproduce the full SegNeXt architecture, but rather to implement a simplified,

SegNeXt-inspired baseline within our pipeline. The model follows a standard encoder-decoder design, emphasizing convolutional feature extraction and multi-scale context aggregation. This lightweight setup makes it feasible under project constraints while remaining relevant for scientific forgery localization. Additionally, we evaluate a pretrained variant using a `convnext_tiny` backbone from `timm` to assess the impact of stronger pretrained visual features. For more implementation details, see GitHub [19].

2.7 Mask post-processing

Post-processing steps were critical for performance under the competition metric. In particular, removing small connected components significantly improved the oF1 score by reducing false positives on authentic images. In the baseline method, we employ the following post-processing steps in order: gaussian blur, confidence threshold filtering based on the threshold 0.5, closing with a 5×5 structuring element, opening and binary filling. The gaussian blur, closing, opening and binary filling operation are all implemented through the `scipy.ndimage` library. The PatchMatch implementation runs similar post-processing but first applies filtering of small components in order to reduce false positives.

3 Results

3.1 Main results

At the time of writing, the maximum oF1 score achieved on hold-out test data was 0.458, achieved by Kaggle user *orshanec*. [1] We again emphasize that our architecture cannot be directly compared with this metric due to us being unable to access the Kaggle hold-out test set. We observe best overall performance from our BusterDino method on our internal hold-out test set.

3.1.1 Baseline model results

Our baseline model achieved an oF1 score of 0.5007, with mean scores of 0.9958 and 0.0722 on authentic and forged images respectively. We observed that it was reluctant to make predictions, predicting few false positives but also few true positives. In general, although we view model performance primarily through the lens of the oF1 metric, high recall is desirable for forgery detection. We theorize that during training, the loss of predicting incorrectly on authentic images punishes the model so much that it instead rarely makes a prediction. The performance measure discussed earlier supports this approach. As we are using a stratified holdout set with 238 authentic and 275 forged, the semi-balance in classes

helps, but if the holdout were to be dominated by forged samples, it would perform worse. Furthermore, although the simplicity of the decoder gives a rapid runtime for this method, we believe that this held back the results for forged predictions.

3.1.2 Deep Cross-Scale PatchMatch results

The PatchMatch architecture achieved competitive performance, reaching mean scores of 0.613 and 0.340 on authentic and forged images respectively, giving a combined score of 0.466. We observe that the true positive rate is much improved compared to the baseline, at the cost of a higher false positive rate. However, this architecture had much higher runtime and memory consumption compared to both our baseline method and BusterDino. The runtime is particularly dependent on the number of PatchMatch iterations which are performed. This module iteratively executes propagation and evaluation steps "several times" [9, p. 7]. While the original paper does not specify the number of iterations, we found that 24 iterations produced stable results. Using this amount of iterations, PatchMatch on a single image takes $\sim 4s$ on an NVIDIA RTX 4070 TI Super GPU, while other components of the network take less than 0.1s to run.

The results of this method are highly dependent on mask post-processing. In particular, it is crucial to perform filtering of small objects, as the architecture generated many small false positive areas on authentic images, automatically giving an oF1 score of 0 in these cases. We theorize that this tendency to predict false positives is because smaller groups of pixels are more likely to be similar to each other "by accident" than larger groups of pixels. The best version of the pipeline on validation data filters out connected components with area below 920 pixels. Although this means accepting that the model will never be able to deal with very small copy-move forgeries, it led to strong performance on average due to the removal of many false positives.

3.1.3 BusterDino results

BusterDino achieved the strongest holdout oF1 among the three architectures. Results are summarised in Table 1.

Metric	Value
Holdout oF1	0.5138
Authentic mean pixel f1	0.8361
Forged mean pixel f1	0.3292
Forged empty rate	31.6%
Forged precision (non-empty)*	0.719
Forged mean pixel f1 (non-empty)*	0.495

Table 1. BusterDino results. *Non-empty metrics computed on validation forged images with a non-empty prediction.

The model remained conservative with a forged empty rate of 31.6%, but produced high-quality masks when predictions were made. We attribute this to the two-branch structure. When the manipulation-detection and copy-detection branches agree, the fusion head becomes confident. When they diverge, it tends toward an empty prediction. The target masks are typically stronger than source masks, preserving conservatism while still producing accurate forged mask localisation. The added branch complexity also made BusterDino less dependent on aggressive post-processing, as the model better distinguishes noise from genuine small forgeries. See Figures B.1–B.3 in the appendix for probability distributions by size bucket and example predictions.

BusterDino is likely undertrained given the hard epoch cutoff. A characteristic pattern emerged during stage-wise training that validation oF1 initially decreased in stages 2 and 3 as the model began predicting more forged pixels, reducing authentic F1. However, inspection of the predictions showed that forged mask quality improved substantially in later checkpoints, yielding a stronger balanced model overall. The most balanced version was the one chosen for the results, trained across 40 epochs (20/10/10 per stage, $\approx 1h$ on an RTX 4080 Super).

3.2 Comparison Between DINO and SegNeXt-inspired Models

	DINO baseline	SegNeXt-inspired	SegNeXt-family
Encoder	ViT-B/14	custom	convnext_tiny
Pretr.	Yes	No	Yes
Train	900	500	900
LR	3e-5	3e-5	3e-5
Ep.	8	5	12
Best Ep.	8	1	10
Val F1	0.1850	0.0838	0.1827

Table 2. Best configurations for the compared model families

Overall, the DINO-based baseline consistently achieved the highest validation performance across

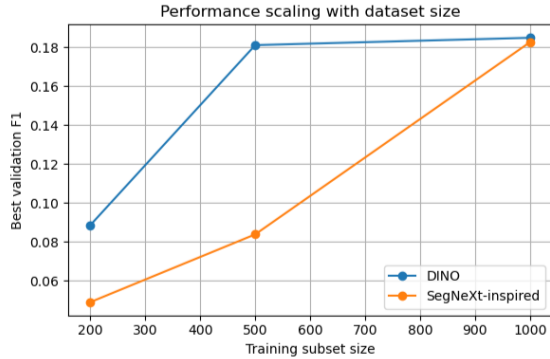


Figure 1. Validation F1 versus dataset size. The DINO baseline performs better at small dataset sizes, while the pretrained SegNeXt-family variant approaches it at the largest setting.

all tested configurations. The pretrained SegNeXt-family model approached this performance, but required more training epochs and relied heavily on pretrained representations. A learning rate of 3×10^{-5} gave substantially stronger validation performance than 10^{-5} for both pretrained methods.

We also attempted using a custom SegNeXt-inspired model without pretraining, but this performed substantially worse, with best validation F1 values remaining below 0.09 across all tested settings. This custom model often reached its best validation F1 already at the first epoch, which suggests unstable optimization and weak feature learning without pretraining, indicating the importance of pretrained visual representations.

4 Discussion

Transfer learning. Transfer learning played a central role in our experiments, as we heavily rely on the strength of the DINOv2 pretrained features. The pretrained SegNeXt-family model approached the DINO baseline, while models trained from scratch performed substantially worse. This suggests that, under limited data, representation quality is a primary bottleneck, and strong pretrained encoders are crucial for effective CMFD.

Loss function misalignment. Although cross-entropy loss correctly penalizes incorrect pixel-level predictions, it is not fully aligned with the strict conditions that the oF1 metric expects. For instance, if a small area of forged pixels is predicted on an authentic image, the impact on the loss is not large, but this makes the oF1 metric 0. We observed that validation loss could decrease while oF1 performance worsened, indicating a mismatch between the training objective and evaluation metric. A promising direction for future work is to design

a differentiable loss more closely aligned with the oF1 objective. As surrogates, we did experiment with employing loss penalties for any forged pixels predicted on authentic images, but a more nuanced alternative would be interesting to experiment with. For both BusterDino and Deep Dino Cross-Scale PatchMatch, adding dice loss in a weighted sum with BCE improved performance greatly compared to regular BCE. This is an example of a problem-specific loss function that accounts for both pixel overlap and binary classification accuracy.

Post-processing. Because of the loss function mismatch, model performance is highly sensitive to post-processing, making it an important direction for further optimization in future work. For instance, filtering out small predicted patches led to much better results, as the oF1 metric also penalizes extra component predictions on forged images. Another pitfall is overly aggressive morphological processing, which can merge separate components into one large connected region.

Failure modes. A major failure mode observed across all models was false positive predictions on authentic images. Due to the strict evaluation metric, even a small predicted region results in an oF1 score of zero. This makes the task highly sensitive to small prediction artifacts.

Dataset. We attribute part of the difficulty to dataset characteristics. Authentic scientific images often contain repeated or visually similar structures, increasing the risk of false positives. This is especially challenging for PatchMatch, which relies on finding regions with high feature similarity.

Li et al. train on synthetic MS COCO copy-move data and evaluate on photo-realistic datasets such as CASIA CMFD, CoMoFoD and CMH [20–23]. These datasets generally contain varied objects and backgrounds, unlike many scientific images, which may have dark or uniform regions. As a result, pixel-wise similarities can produce false positives. Our PatchMatch architecture performed better when the copy-move operation was direct and not heavily post-processed.

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A Implementation

Our implementation is available on GitHub [24].

B BusterDino Diagnostics

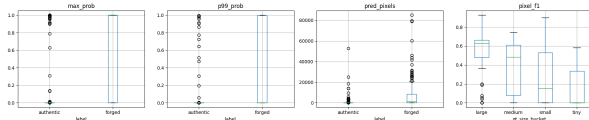


Figure B.1. BusterDino validation distributions. Left to right: maximum predicted probability per image (authentic vs. forged), 99th-percentile probability, total predicted pixels, and pixel F1 by ground-truth mask size bucket. Forged images show bimodal behaviour — the model either fires confidently or produces an empty prediction. F1 degrades for smaller forgeries, consistent with the size-bucket summary in the main text.

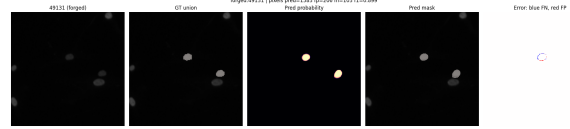


Figure B.2. BusterDino successful prediction on a microscopy image (case 49131, $F1=0.899$). From left: forged image, ground-truth union mask, predicted probability map, predicted binary mask, and error overlay (blue = false negative, red = false positive). The model correctly localises both the source and target regions with minimal error.

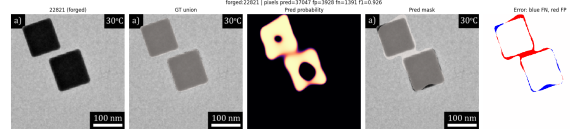


Figure B.3. BusterDino prediction on a nanoparticle image (case 22821, $F1=0.526$). The model identifies the copied rectangular region; residual errors at the edges are visible in the error overlay. This example illustrates the challenge of precise boundary localisation on high-contrast scientific images.

C Deep Dino Cross Scale PatchMatch Post-Processing Performance Gap

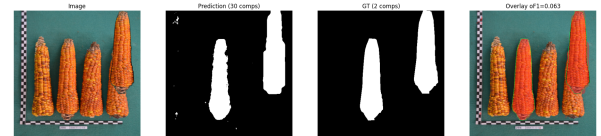


Figure C.1. Deep Dino Cross-Scale PatchMatch on an image using no post-processing, achieving $oF1=0.063$. The model detects the copied object but also generates many small components. This affects the $oF1$ penalty P , leading to an overall low $oF1$ score, even though per-pixel precision is quite strong.

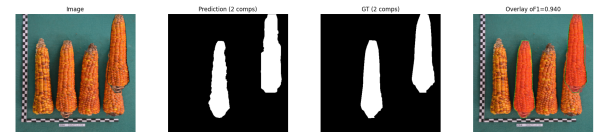


Figure C.2. Deep Dino Cross-Scale PatchMatch on the same figure using post-processing, achieving $oF1=0.948$.

D Dataset examples

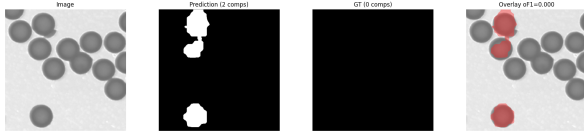


Figure D.1. An authentic image containing very similar objects, causing false positives in the Deep Dino Cross-Scale PatchMatch prediction.

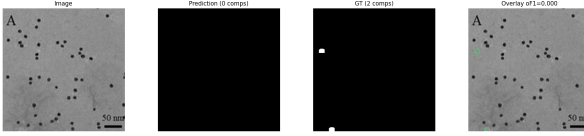


Figure D.2. A forged image with the source being hard to predict due to size. The aggressive minimum component size post-processing performed by some of the models examined in this paper, such as Deep Dino Cross-Scale PatchMatch, mean that they would be unable to detect this forgery.

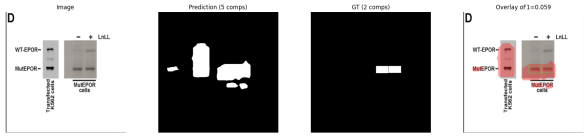


Figure D.3. A scientific image where mask post-processing combines components too aggressively. In addition, this highlights some challenges due to dataset characteristics - the text ("MutEPOR") is predicted as a forgery because it appears in the exact same form twice.